

Exercises

Monads and IO

Exercise 1:

Define the higher-order function *whileG* in which the condition and the operation work over values of type *a*. Its type should be:

`whileG :: (a -> IO Bool) -> (a -> IO a) -> (a -> IO a)`

so that

`whileG cond op x`

has the effect of repeating *op x* while the condition *cond x* is *True*.

Exercise 2:

Using the function *whileG* or otherwise, define an interaction which reads a number, *n*, say, and then reads a further *n* numbers and finally returns their average.

Exercise 3:

Given the type definition

`data Result a = Succeed a | Fail`

show how *Result* can be made into a monadic type.